Presenting Regression Results Using R

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Introduction

Regression analysis is a fundamental statistical technique used in business and research to understand relationships between variables. However, running a regression is only half the battle—presenting the results clearly and accurately is equally important. This handout covers best practices for presenting regression results using R, focusing on clear interpretation, appropriate visualizations, and professional reporting standards.

Understanding Regression Results

The Regression Model

Linear regression examines how one variable (the outcome) depends on other variables (predictors). The basic equation is:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon$$

Where: - Y is the outcome variable - β_0 is the intercept - β_1, β_2 are coefficients (how much Y changes per unit of X) - ϵ is the error term

Key Statistics to Report

Coefficients: How much does Y change when X increases by one unit?

P-values: Is the effect real or due to chance? If $p < 0.05$, the effect is statistically significant.

R^2: What percentage of variation in Y is explained by the model? Range: 0 to 1.

Standard Error: Uncertainty around each coefficient estimate.

Presenting Regression Results in R

Method 1: The Summary Table

The `summary()` function gives basic output. For professional presentation, use the `stargazer` package:

```r
library(tidyverse)
library(stargazer)
library(broom)

# Load example data
data(mtcars)

# Fit regression model
model1 <- lm(mpg ~ wt + cyl, data = mtcars)

# Professional table
stargazer(model1, type = "text",
          title = "Regression Results: Fuel Efficiency",
          summary.stat = c("n", "mean", "sd"))
```

```
Regression Results: Fuel Efficiency
===============================================
                        Dependent variable:
                    ---------------------------
                                mpg
-----------------------------------------------
wt                           -3.191***
                              (0.757)

cyl                          -1.508***
                              (0.415)

Constant                     39.686***
                              (1.715)

-----------------------------------------------
Observations                    32
R2                             0.830
Adjusted R2                    0.819
```

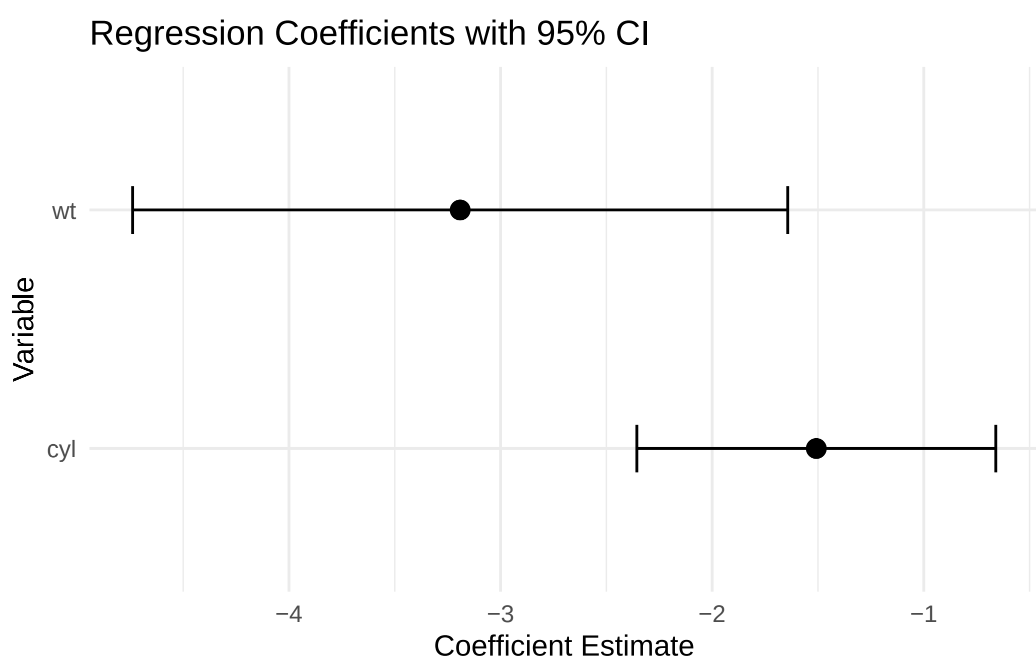
```
Residual Std. Error       2.568 (df = 29)
F Statistic            70.908*** (df = 2; 29)
===============================================
Note:                  *p<0.1; **p<0.05; ***p<0.01
```

Method 2: Visualization

Charts are often more intuitive than tables. Visualize coefficients and their uncertainty:

```r
# Extract results in tidy format
results <- tidy(model1, conf.int = TRUE)

# Plot coefficients
ggplot(results %>% filter(term != "(Intercept)"),
       aes(y = term, x = estimate)) +
  geom_point(size = 3) +
  geom_errorbar(aes(xmin = conf.low, xmax = conf.high),
                width = 0.2) +
  labs(title = "Regression Coefficients with 95% CI",
       x = "Coefficient Estimate",
       y = "Variable") +
  theme_minimal()
```



Method 3: Predicted Values

Show what the model predicts for different input values:

```
# Create scenario: different weights
scenarios <- tibble(wt = c(2, 3, 4, 5), cyl = 6)

# Get predictions
predictions <- predict(model1, newdata = scenarios,
                       se.fit = TRUE)

scenarios %>%
  mutate(
    predicted_mpg = predictions$fit,
    se = predictions$se.fit,
    ci_lower = predicted_mpg - 1.96 * se,
    ci_upper = predicted_mpg + 1.96 * se
  ) %>%
  knitr::kable(digits = 2, caption = "Predicted MPG by Weight and Cylinders")
```

Table 1: Predicted MPG by Weight and Cylinders

wt	cyl	predicted_mpg	se	ci_lower	ci_upper
2	6	24.26	0.97	22.35	26.17
3	6	21.07	0.47	20.15	21.98
4	6	17.88	0.80	16.31	19.44
5	6	14.68	1.48	11.78	17.59

Key Takeaways for Presentation

1. **Know your audience**: Non-technical audiences prefer charts; academic audiences need tables.
2. **Report uncertainty**: Always show confidence intervals or standard errors, not just point estimates.
3. **Check assumptions**: Validate that regression assumptions hold before reporting.
4. **Interpret in context**: Explain what numbers mean for the business question.
5. **Avoid common mistakes**: Don't confuse correlation with causation; acknowledge limitations.

References

Affidavit

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Date: 01 June 2026

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